REVISION 2021

Course Code: 2059 | Semester: 2 | Diploma in Automobile Engineering

# Course Overview

## Objectives

- Hands-on experience with basic automobile hand tools.
- Identification of engine components through dismantling.
- Understanding basic engine experiments and servicing procedures.
- Enhancing practical automotive servicing skills.

## Teaching Scheme

| | |
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| **Category** | Engineering Science |
| **Periods/Week** | 3 (Practical) |
| **Total Periods** | 45 |
| **Credits** | No Credit (Audit/Lab focus) |

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## Official Source Declaration

This lesson is strictly mapped to the **SITTTR Kerala Revision 2021** syllabus for Course Code 2059. All modules, outcomes, and topics are derived from the official curriculum document provided by the State Institute of Technical Teachers' Training and Research.

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## Course Outcomes (COs)

| CO No. | Description | Hours | Cognitive Level |
|---|---|---|---|
| CO1 | Demonstrate the use of different tools used in Automobile workshop | 7 | Applying |
| CO2 | Identify the application of measuring tools and specific tools in Automobile Garage. | 7 | Applying |
| CO3 | Identify various engine components in an Automobile after dismantling. | 14 | Applying |
| CO4 | Apply the procedure of maintenance schedule on two and four-wheeler vehicles. | 14 | Applying |

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## Module 1: Automobile Workshop Tools

Duration: 7 Hours | CO1

### M1.01: Identification of Hand Tools

Automobile servicing requires a variety of specialized hand tools to ensure fasteners are handled without damage and components are removed safely.

- **Spanners:** Open-end, Ring, and Combination spanners. Used for tightening/loosening bolts.
- **Socket Sets:** Used with ratchets for faster operation in confined spaces.
- **Wrenches:** Adjustable wrenches, Pipe wrenches, and Torque wrenches.
- **Pliers:** Combination pliers, Nose pliers, and Circlip pliers (Internal/External).
- **Hammers:** Ball peen, Cross peen, and Soft-faced (Mallet) hammers.
- **Screwdrivers:** Flat head, Phillips head, and Torx drivers.

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### M1.02: Special Tools and Competency

Special tools are designed for specific tasks that standard tools cannot perform efficiently.

- **Torque Wrench:** Essential for tightening cylinder head bolts to specific manufacturer values to prevent warping.
- **Feeler Gauge:** Used to measure small gaps, such as valve clearance (tappet setting) or spark plug gaps.
- **Pullers:** Gear pullers and bearing pullers used to remove components pressed onto shafts.

## Module 2: Measuring Tools & Engine Calculations

Duration: 7 Hours | CO2

### M2.01 & M2.02: Bore and Stroke Measurement

Precision measurement is vital for determining engine wear and calculating displacement.

- **Bore Gauge:** A telescopic tool used to measure the internal diameter (Bore) of a cylinder. It helps identify "ovality" and "taper" wear.
- **Stroke Length:** The distance the piston travels from Top Dead Center (TDC) to Bottom Dead Center (BDC). It is exactly twice the **Crank Radius (r)**.

**Stroke (L) = 2 × Crank Radius (r)**

### M2.03 & M2.04: Swept Volume and Compression Ratio

These calculations define the engine's capacity and thermal efficiency.

**Swept Volume ($V_s$):** The volume displaced by the piston during one stroke.
Formula: $V_s = (\pi/4) \times D^2 \times L$
Where D = Bore diameter, L = Stroke length.

**Compression Ratio (CR):** The ratio of total cylinder volume to clearance volume.
Formula: $CR = (V_s + V_c) / V_c$
Where $V_c$ = Clearance Volume (volume above piston at TDC).

## Formula Bank & Interactive Calculator

Bore (D) in mm:

e.g. 70

Stroke (L) in mm:

e.g. 80

Clearance Vol (Vc) in cc:

e.g. 40

Calculate

## Visual Library

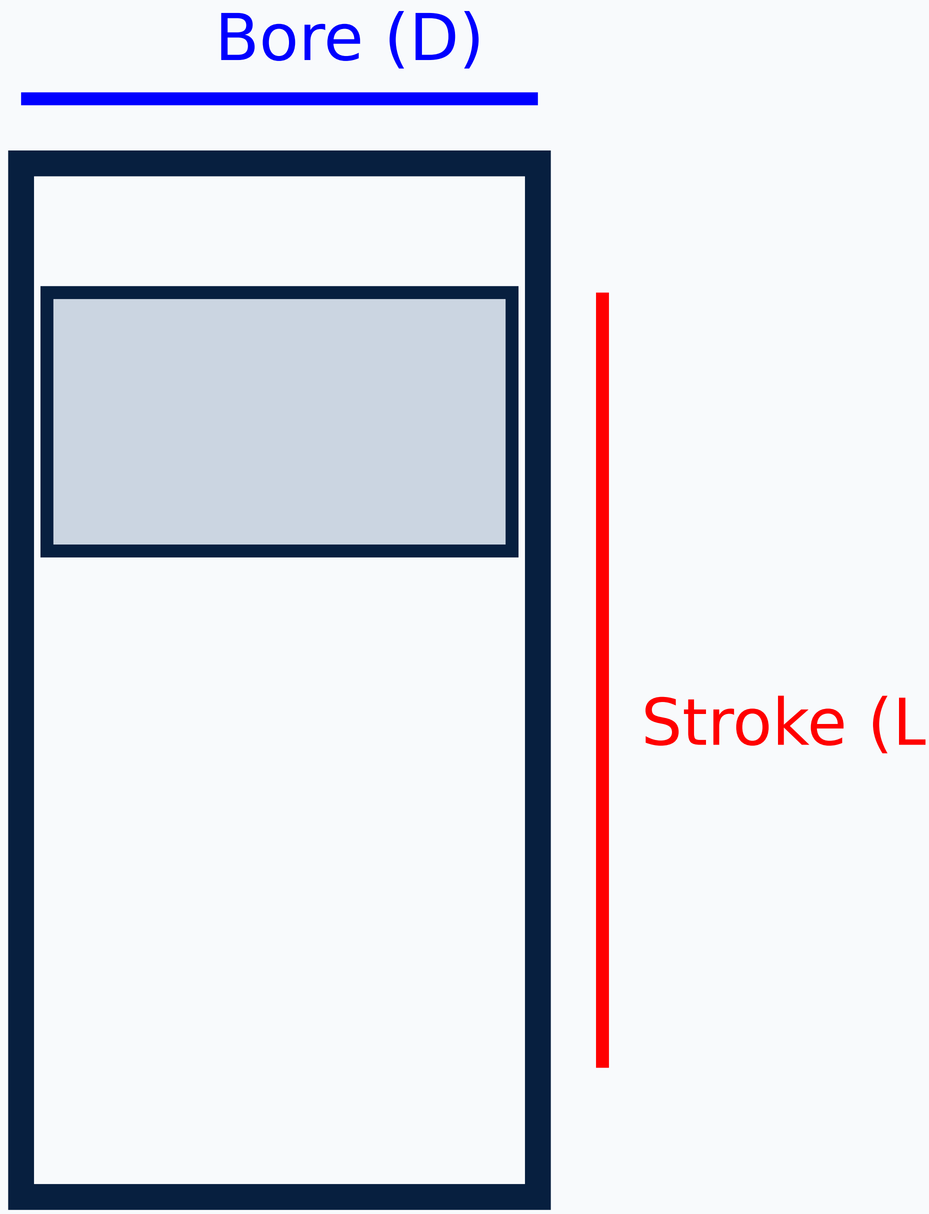


Figure 1: Basic Engine Cylinder Dimensions (Bore and Stroke)

# Module 3: Engine Components Identification

Duration: 14 Hours | CO3

This module involves the actual dismantling of an engine to identify internal parts.

| **Component** | **Function / Key Details** |
|---|---|
| Cylinder Block | The main structure of the engine; contains cylinders. |
| Cylinder Liners | **Dry Liner:** No direct contact with coolant. **Wet Liner:** Direct contact with coolant. |
| Piston & Rings | Transmits force to con-rod. Includes **Compression rings** (seal gas) and **Oil rings** (scrape oil). |
| Gudgeon Pin | Connects the piston to the small end of the connecting rod. |
| Crankshaft | Converts reciprocating motion of piston into rotary motion. |
| Camshaft | Controls the opening and closing of valves. |

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# Module 4: Maintenance & Servicing

Duration: 14 Hours | CO4

**M4.01 - M4.04: Fluids and General Maintenance**

**M4.05 - M4.07: Spark Plug and Injector Testing**

## Practice Questions & Answers

1. What is the difference between a wet liner and a dry liner?

2. Define Compression Ratio.

3. Why is a torque wrench used for cylinder head bolts?

## Practice Model Lab Exam

### Time: 3 Hours | Max Marks: 100

**Part A: Identification (20 Marks)**
Identify 5 engine components placed on the table and state their functions.

**Part B: Measurement (30 Marks)**
Measure the bore and stroke of the given engine. Calculate the swept volume.

**Part C: Maintenance Task (50 Marks)**

Perform spark plug cleaning and gap setting OR Demonstrate the procedure for engine oil level check and jacking up a vehicle.

## Quick Revision

- **Stroke = 2 × Crank Radius.**
- **Swept Volume** is the capacity of one cylinder.
- **Feeler Gauge** is for gaps; **Bore Gauge** is for internal diameters.
- **Compression Rings** prevent blow-by; **Oil Rings** control lubrication.
- Always use **Safety Stands** after jacking up a vehicle.

## Glossary

| | |
|---|---|
| **TDC** | Top Dead Center (Highest point of piston travel). |
| **BDC** | Bottom Dead Center (Lowest point of piston travel). |
| **Ovality** | Difference in cylinder diameter measured in two perpendicular directions. |
| **Viscosity** | Resistance of oil to flow. |

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1. **[illegible]:** [illegible], [illegible].
2. **[illegible]:** [illegible] (Bore), [illegible] (Stroke) [illegible].
3. **[illegible]:** [illegible], [illegible], [illegible].
4. **[illegible]:** [illegible], [illegible], [illegible].

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Reference: Basic Automobile Engineering - C P Nakra | Practical Automobile Engineering – Malhotra